

EDUCATION

PhD Hypergraph versus Graph Metrics

2024 - present

University of Amsterdam

Zaragoza, Spain

Study on higher-order interactions in the emergence of complex dynamical behaviour. My main focus is two-fold; extracting significant time- and space-like interactions in multivariate systems, and relating macroscopic structural properties to system dynamics.

MSc Computational Physics of Complex Systems

2022 - 2024

University of Amsterdam

Amsterdam, Netherlands

Study on the emergent phenomena in complex many-body and nonlinear systems, using a wide range of computational methods. Thesis: Exploring the Helmholtz Free Energy as a Potential Characteristic for Identifying Macroscopic Structural Properties of Simplicial Complexes

MSc minor Space Physics (completed)

2022 - 2023

The Arctic University of Norway

Svalbard, Norway

Plasma- and statistical physics specialised to interaction between solar activity and the Earth's magnetosphere in the High Arctic atmosphere.

BSc Physics and Astronomy (completed)

2018 - 2021

University of Amsterdam/VU University

Amsterdam, Netherlands

Study with focus on classical physics, modern physics and astronomy, aiding the development of analytical, critical and mathematical thinking skills

BSc Medical Natural Sciences (completed)

2017 - 2021

VU University

Amsterdam, Netherlands

Study combining exact sciences and the biomedical field, developing knowledge on medical innovations and (patho)physiology to serve as a bridge connecting fundamental research with clinical practice.

BSc Molecular Life Sciences

2016 - 2017

Radboud University

Nijmegen, Netherlands

Interdisciplinary study of Science, Mathematics and Computation, strongly connected to the Radboudumc Research Institute for Medical Innovation. After a year of study, I came to the realization that I yearned for a deeper understanding of physical theory.

WORKSHOPS, SCHOOLS AND CONFERENCES

International School and Conference on Network Science

The Network Science Society

June 2025

Maastricht, Netherlands

The International School and Conference on Network Science (NetSci 2025)

High Order Interactions BrainMod

Institute of Advanced Study

May 2025

Amsterdam, Netherlands

Workshop on Brain Modeling and High-Order Interactions with the use of Information Theory

Brain Simulation workshop

The Human Brain Project

June 2024

Bilbao, Spain

5-day workshop on new techniques in neural network simulation by EBRAINS, Europe's Research Infrastructure for Brain Research.

EXPERIENCE

High Order Interactions BrainMod

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Research Internship*Bilbao Centre of Applied Mathematics*

Internship at the Bilbao Centre of Applied Mathematics exploring Bayesian Inference, Information Homology, Free Energy, and Topological Data Analysis of the brain.

Feb 2024 – Apr 2024, Jun 2024

*Bilbao, Spain***Highschool teacher***Huygens College Amsterdam*

High school teacher in Physics and Mathematics at Huygens College Amsterdam

Sep 2023 – Feb 2024

*Amsterdam, Netherlands***Elementary tutor***Bijles Academie*

Individual tutoring in mathematics for children in elementary school.

Sep 2023 – Feb 2024

*Amsterdam, Netherlands***Personal Mentor/Teaching Assistant***University of Amsterdam*

For two years, I guided students with extreme study delays due to personal issues and mental health problems.

Sep 2020 – Sep 2022

*Amsterdam, Netherlands***Nutritionist Patient Care***Canisius Wilhelmina Hospital*

During my tenure at the hospital, I dedicated myself to delivering comprehensive patient care, with a primary focus on monitoring and tailoring nutritional intake to improve the physical and mental recovery process.

Sep 2015 – Sep 2017

Nijmegen, Netherlands

EXTRA-CURRICULAR ACTIVITIES**AE-E Aesthetics, Entrepreneurship**

2016 - present

Musical endeavor dedicated to fostering the growth of the Dutch post-punk scene. High activity during 2020-2021, including a debut-tour of 42 shows travelling across the Netherlands, the United Kingdom, Belgium, and Germany.

Minor Media Studies Research Master - Film Studies (completed)

2021 - 2022

MSc level minor in Film and Documentary at the University of Amsterdam to improve my skills in science communication, final grade 8.0.

LANGUAGES**Spanish proficiency, NLQF 1 / EQF 1***Issued Nov 2024***English proficiency, ETS TOEFL 110/120***Issued Feb 2022***French proficiency, NLQF 4+ / EQF 4***Issued Sep 2016*

TECHNICAL SKILLS**Languages:** Python, Matlab, Mathematica, R**Packages and DataCamp courses:** Shell, Bash, Pandas, NetworkX, Gudhi, Neuron**Supercomputing:** Use of the Dutch supercomputer Snellius and the European Fenix research infrastructure

CERTIFICATIONS**Suicideprevention Gatekeeper, 113 Suicide Prevention***Issued Sep 2022***First Aid Essentials, Red Cross***Issued Oct 2023***DataCamp courses***Issued Oct 2023*